

100xEngineers

The Machine Learning Path

Machine Learning → Deep Learning → LLMs & Generative AI

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Phase 0: Prerequisites

Before touching ML, you need a solid foundation in Python and math. Don't skip this, everything builds on it.

Python Programming

- **Python for Everybody** (Coursera, Dr. Chuck) Free and great course coursera.org
- **Bigbinary Python Tutorial** Free. Quick reference and interactive exercises. [bigbinary](https://bigbinary.com)
- **Practice:** Build 3–5 small projects (calculator, file organizer, web scraper) to get comfortable.

Mathematics for ML

- **3Blue1Brown — Essence of Linear Algebra** (YouTube) — Free. Best visual explanation of linear algebra ever made. [YouTube Playlist](https://www.youtube.com/playlist?list=PLZHQObOWTQDPBV3I_r-pQjsI_r0YxNjV-q)
- **3Blue1Brown — Essence of Calculus** (YouTube) — Free. Visual-first approach for derivatives, integrals, chain rule. [YouTube Playlist](https://www.youtube.com/playlist?list=PLZHQObOWTQDPBV3I_r-pQjsI_r0YxNjV-q)
- **Khan Academy — Statistics & Probability** — Free. Distributions, Bayes' theorem, hypothesis testing. khanacademy.org
- **Mathematics for ML Specialization** (Coursera, Imperial College London) — Free to audit. coursera.org

Data Handling

- **Kaggle Learn: Pandas** — Free. Short, interactive micro-course. kaggle.com/learn/pandas
- **Kaggle Learn: Data Visualization** — Free. Matplotlib and Seaborn basics. kaggle.com/learn/data-visualization

Phase 1: Machine Learning Foundations

This is where you learn what ML actually is — the algorithms, the intuition, and the workflow.

Primary Course (Pick One)

- **Andrew Ng's Machine Learning Specialization** (Coursera, DeepLearning.AI + Stanford) — Free to audit. The most recommended starting point in the world. Covers supervised learning, unsupervised learning, and recommender systems. ~50–100 hours. coursera.org
- **Google's Machine Learning Crash Course** — Free. Faster-paced alternative with TensorFlow exercises. developers.google.com

Supplementary Learning

- **StatQuest with Josh Starmer** (YouTube) — Free. Visual, step-by-step algorithm breakdowns. One of the best ML explainers. youtube.com/@statquest
- **Sentdex** (YouTube) — Free. Python ML tutorials, project-based approach. youtube.com/@sentdex

Books

- **An Introduction to Statistical Learning (ISLR)** — Free PDF. Best introductory ML theory textbook. statlearning.com
- **Hands-On ML with Scikit-Learn, Keras, and TensorFlow** (Aurélien Géron) — Paid (~\$50). The practitioner's bible.

Practice

- **Kaggle** — Start with Titanic and House Prices competitions. Build 3–5 end-to-end projects. kaggle.com

Phase 2: Deep Learning & Neural Networks

From classical ML into neural networks, CNNs, RNNs, and modern architectures.

Primary Courses

- **Deep Learning Specialization** (Coursera, Andrew Ng / DeepLearning.AI) — Free to audit. Five courses: neural network fundamentals, hyperparameter tuning, regularization, CNNs, and sequence models. ~80 hours. coursera.org
- **Practical Deep Learning for Coders** (fast.ai, Jeremy Howard) — Free. Top-down, code-first approach. Build working models from day one. Uses PyTorch/fastai. course.fast.ai

The Must-Watch Series

- **Andrej Karpathy — Neural Networks: Zero to Hero** (YouTube) — Free. A masterclass. Builds neural networks from scratch: backpropagation (micrograd), MLPs, BatchNorm, and culminating in building a GPT-like transformer. Arguably the single best resource for deeply understanding how neural networks work. karpathy.ai/zero-to-hero

Supplementary

- **MIT 6.S191: Introduction to Deep Learning** — Free. Fast-paced MIT course with hands-on labs. introtodeeplearning.com
- **Stanford CS231n: CNNs for Visual Recognition** — Free lectures on YouTube. Deep dive into CNNs. cs231n.stanford.edu

- **3Blue1Brown — Neural Networks** (YouTube) — Free. 4-part visual series for intuition. [YouTube Playlist](#)

Frameworks

- **PyTorch** — Most popular in research. Karpathy, fast.ai, and most papers use it. Recommended primary. pytorch.org/tutorials
- **TensorFlow / Keras** — Widely used in production/enterprise. Andrew Ng's courses use it. Learn as secondary. tensorflow.org/tutorials

Books

- **Deep Learning** (Goodfellow, Bengio, Courville) — Free online. The definitive reference textbook. deeplearningbook.org
 - **Dive into Deep Learning (d2l.ai)** — Free online. Interactive, code-heavy textbook. d2l.ai
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Phase 3: Transformers & NLP

Transformers are the architecture behind every modern LLM. This phase bridges deep learning and generative AI.

Core Learning

- **Andrej Karpathy — “Let’s build GPT”** (YouTube) — Free. Build a GPT from scratch. [YouTube](#)
- **Hugging Face NLP Course** — Free. Tokenization, transformer architectures, fine-tuning. huggingface.co/learn/nlp-course
- **Stanford CS224N: NLP with Deep Learning** — Free lectures on YouTube. The gold standard NLP course. web.stanford.edu/class/cs224n

Key Paper

- **“Attention Is All You Need”** (Vaswani et al., 2017) — The original transformer paper. arxiv.org

Supplementary

- **Jay Alammar’s Blog** — Free. Best visual explanations of transformers. “The Illustrated Transformer” is essential. jalammar.github.io
 - **Lilian Weng’s Blog** — Free. In-depth technical posts on attention, transformers, prompt engineering. lilianweng.github.io
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Phase 4: LLMs & Generative AI

Understanding, building with, and deploying large language models and generative AI systems.

Foundational Understanding

- **Andrej Karpathy — “Intro to Large Language Models”** (YouTube) — Free. 1-hour overview. [YouTube](#)
- **Generative AI with LLMs** (Coursera, DeepLearning.AI + AWS) — Free to audit. LLM lifecycle: transformer architecture, training, fine-tuning (PEFT/LoRA), RLHF, deployment. [coursera.org](#)
- **Generative AI for Beginners** (Microsoft, GitHub) — Free. 21 lessons on GenAI, prompt engineering, RAG, responsible AI. [github.com/microsoft](#)

LLM Engineering & Building Applications

- **Cohere’s LLM University** — Free. Transformer architecture, embeddings, semantic search, RAG, AI agents. [cohere.com/llmu](#)
- **DeepLearning.AI Short Courses** — Free. Dozens of 1-hour focused courses: LangChain, RAG, fine-tuning, vector databases. [deeplearning.ai/short-courses](#)
- **LLM Twin Course** (Decoding ML, GitHub) — Free. 12 lessons on building a production RAG system. [github.com/decodingml](#)

Key Topics to Master

- **Prompt Engineering:** Few-shot, chain-of-thought, system prompts, structured outputs.
- **RAG:** Vector databases (Pinecone, Chroma, FAISS), embedding models, chunking strategies.
- **Fine-tuning:** LoRA, QLoRA, PEFT methods for adapting models to specific tasks.
- **AI Agents:** Function calling, tool use, multi-agent systems, LangChain, CrewAI.
- **Evaluation:** Benchmarking, human evaluation, automated metrics for generative outputs.

Advanced / Going Deeper

- **Maxime Labonne — LLM Course** (GitHub) — Free. Architecture to deployment. [github.com/mlabonne](#)
- **“AI Engineering” by Chip Huyen** — Paid book. MLOps, inference, data-centric AI.
- **“The LLM Engineer’s Handbook”** (Iusztin & Labonne) — Paid. Vector search, prompt chaining, evals, RAG.

Stay Current

- **Andrej Karpathy** on YouTube and X/Twitter — [karpathy.ai](#)
- **The Batch Newsletter** (Andrew Ng) — [deeplearning.ai/the-batch](#)

- **Hugging Face Blog** — huggingface.co/blog
- **Simon Willison's Blog** — Practical LLM insights — simonwillison.net

Summary: The Sequenced Path

Phase	Focus	Duration	Key Resource
0	Prerequisites (Python + Math)	2–4 weeks	3Blue1Brown + Python for Everybody
1	Machine Learning Foundations	6–10 weeks	Andrew Ng's ML Specialization + StatQuest
2	Deep Learning & Neural Networks	8–12 weeks	DL Specialization + Karpathy Zero to Hero
3	Transformers & NLP	4–6 weeks	Karpathy's GPT video + Hugging Face Course
4	LLMs & Generative AI	6–10 weeks	GenAI with LLMs + DL.AI Short Courses

Total estimated time: 6–10 months (studying 8–12 hours/week)

Principles for the Journey

Build constantly. Every phase should include projects. Reading and watching without building is an illusion of learning. Implement papers, compete on Kaggle, build tools for yourself.

Go depth-first on fundamentals. Karpathy's Zero to Hero series and Andrew Ng's specializations build genuine understanding, not surface familiarity.

Don't chase every new tool. The frameworks and APIs will change. The math, the intuition for how models learn, the ability to debug training, runs that last.

Use the community. Kaggle, Hugging Face, Discord servers, fast.ai forums, and r/MachineLearning are valuable for getting unstuck and staying motivated.